FIPA — towards a standard for software agents

P D O'Brien and R C Nicol

Software agent technology will have a significant impact on the shape of the global information society in the next millennium. This branch of systems engineering is rapidly becoming a viable and exploitable technology where BT is well placed as both a potential user and provider of agent-based services and products.

The highly interactive nature of multi-agent systems points to the need for consensus on agent interfaces in order to support interoperability between different agent systems. The completion and adoption of such a standard is a prerequisite to the commercialisation and successful exploitation of agent technology.

This paper describes FIPA (Foundation for Intelligent Physical Agents) and its organisation. It provides an overview and guide to the FIPA97 specification. It discusses how FIPA relates to other agent standards activities and concludes with FIPA's plans for 1998.

1. Introduction

An agent is a piece of software capable of acting intelligently on behalf of a user or users, in order to accomplish a task. Agents, like humans, co-operate so that a society of agents can combine their abilities to resolve problems. Agent technology is currently being applied to a number of application areas including network management [1], information filtering [2, 3], user modelling [4, 5], and business process management [6]. Due to the collaborative nature of agent systems, an agent standard represents a key requirement for the commercialisation of agent technology.

1.1 Motivation for an agent standard

The need for an agent standard can be illustrated using a simple travel booking example. A customer requests their personal agent to find a particular type of holiday, within a specific price range, with a choice of airports, for a particular set of dates. The personal agent searches for all available agents offering retail travel services, requests each to provide information on holidays that match its criteria, ranks the suitable holidays based on its knowledge of the user's preferences, and finally reports back to its user. The user selects their preferred holiday and authorises the agent to agree a contract and pay the required deposit [7].

The scenario contains two types of agent — customer agents which solely represent the interests of a customer, and travel retail agents offering retail travel services and representing the interests of travel retailers. Each agent would be owned by and represent a different party. Similarly, each agent could have been developed by different companies, at different times and using different underlying agent technologies [8, 9]

For such a scenario to work there needs to be a minimum set of agent standards which provide:

- a commonly agreed means by which agents can communicate with each other so that they can exchange information, negotiate for services, or delegate tasks,
- facilities whereby agents can locate each other (i.e. directory facilities),
- an environment which is secure and trusted where agents can operate and exchange confidential messages,
- a unique way of agent identification (i.e. globally unique names),
- a means of accessing back-end or legacy systems,
- a means of interacting with users,
- if necessary, a means of migrating from one platform to another.

An agent standard will also provide confidence for the development of an agent-component market. Early adopters of new technologies tend to be wary where there is no consensus across appropriate industries on a new technology. An agent standard will provide added confidence to potential adopters of this technology. The standardisation process also shifts the debate from longer term research issues to the practicalities of realising commercial agent systems. Organisations, such as the Foundation for Intelligent Physical Agents (FIPA), allow focused collaboration in addressing the key challenges facing commercial agent developers as they bring this technology to market.

1.2 FIPA

FIPA was established in December 1995 as a non-profit organisation for producing specifications for open agent interfaces. Its primary purpose was to create consensus between different organisations involved in agent technology, and so initiate the agent standardisation process. Currently it has over 44 international member organisations including Alcatel, BT, NTT, IBM, HP, France Telecom, Siemens, Hitachi, and Toshiba. FIPA, as an open organisation, operates through a consensual process where commercial, rather than academic, interests have been well represented.

It is a representative body for commercial agent interests and as such is the appropriate body for agreeing common agent specifications. FIPA is defining consortia standards[1] as a means of promoting emerging agent applications, services and products. Already there are indications that agent component products are emerging based on the initial FIPA specifications [2].

1.3 The FIPA specifications

The FIPA specifications represent the first step towards an agent standard. They do not attempt to prescribe the internal architecture of agents nor how they should be implemented, but they do specify the interfaces necessary to support interoperability between agent systems. FIPA97 (issued in October 1997) [10—12] is the first output from FIPA covering part of the requirements for an agent standard. FIPA98 is currently being produced for release in October 1998 which covers additional areas.

Four areas of standardisation have been identified by FIPA (see Fig 1).

[1] Standards that are produced openly but agreed only by member companies.
[2] At the FIPA Osaka meeting (April 1998) GMD-Fokus announced their intention of producing a FIPA compliant version of the Grasshopper mobile agent platform by the end of 1998.

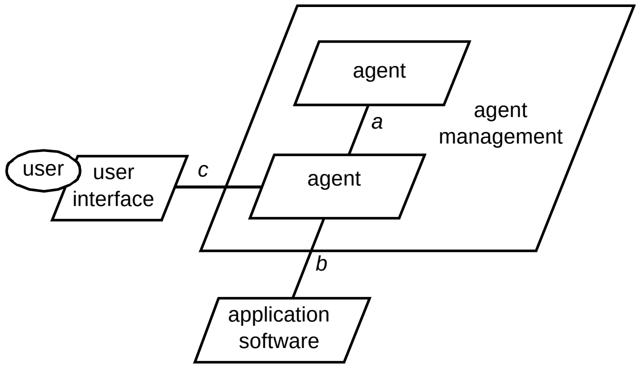

Fig 1 Key agent interfaces.

- Agent communication (interface *a* in Fig 1)

  This facilitates communication between agents, and could support agent interactions such as negotiation, co-operation and information exchange. An agent communication language (ACL), FIPA ACL, has been specified by FIPA to support this interface.

- Agent management

  This includes interfaces to the facilities necessary to support the location and creation of agents, the communication between agents as well as facilities for security and mobility. FIPA97 part 1 specifies facilities for agent management, although the mobility and security issues will not be covered until FIPA98.

- Agent/software integration (interface *b*)

  This interface supports interaction between an agent and other non-agent software. This could include back-end or legacy systems, or hardware drivers (e.g. in robotics).

- Human/agent interaction (interface *c*)

  This covers how agents can interact with the human users and/or other agents manipulating user-related information such as user models. This interface is being specified as part of FIPA's 1998 work-plan and so there will be no specification issued until FIPA98 is issued.

The FIPA97 specification [10—12] contains three normative specifications (parts 1—3), and four informative specifications (parts 4—7) (see Table 1). Conformant agent systems must follow parts 1—3. The informative parts provided guidance during the development of the normative specifications and are intended to support developers when interpreting the normative specifications. These are based on application scenarios as a means of illustrating the application of agents and the use of the FIPA specification.

Table 1 Structure of the FIPA97 specification.

FIPA97 specification			
Normative		Information	
Part 1	Agent management	Part 4	Personal travel assistance
Part 2	Agent communication	Part 5	Personal assistant
Part 3	Agent software integration	Part 6	Audio/visual entertainment
		Part 7	Network management

2. Agent communication

The ability of agents to co-operate requires a language powerful enough to express an agent's **beliefs** and **desires**. FIPA assumes that agents need to interact at a semantically rich level of discourse so as to communicate knowledgeable and meaningful statements about the world. This is a key feature which differentiates agents from other encapsulated software entities such as CORBA objects.

2.1 Background to FIPA ACL

In defining an agent communication language, FIPA had to consider whether or not to adopt the existing commonly used agent communication language, KQML (knowledge querying and communication language).

KQML [13] is widely used in the agent community, and was developed specifically as part of the KSE (Knowledge Sharing Effort) consortium [14]. As a standard, KQML has a number of shortcomings [15]. A significant drawback of KQML is the lack of a universally agreed formal semantics. The semantics of KQML is informally defined in natural language and is therefore open to interpretation by each developer who uses the language. Admittedly, there has been subsequent work, particularly Labrou [16, 17] where a formal semantics has been defined for KQML, but it is unclear how widely this has been adopted by KQML users. Additionally, the KQML language has been extended and now represents an unwieldy 36 speech act performatives.

Other criticisms of KQML [18] included its lack of any commissives (e.g. agreeing to perform an action) and its structural approach to agent interaction (i.e. certain communicative acts are strongly linked to others, for example, ASK-IF with TELL/UNTELL).

FIPA finally agreed that the basis for its ACL was to be an alternative language, ARCOL [19—22]. ARCOL is a semantically well-defined, highly expressive agent communication language which provides a set of primitives from which other agent messages can be created. The proponents of ARCOL successfully argued that ARCOL was more expressive and more formally defined than KQML. Despite ARCOL being the basis for FIPA ACL, the final version of the language benefited from numerous discussions where both ARCOL and KQML viewpoints were well represented. The final specification (FIPA97 part 2) combines the formality of ARCOL with the user friendliness and usability of KQML.

2.2 FIPA ACL agent message

The FIPA ACL specifies communication between agents and has an associated formal semantics. It can be viewed as consisting of five levels :

1 protocol — defines the social rules for structuring the dialogue between agents,

2 communicative act (CA) — defines the type of communication being performed, for example, request is a type of CA.

3 messaging — defines meta-information about the agent message including identity of sending agent, receiving agent, context,

4 content language — defines the grammar and associated semantics for expressing the content of a message, an example possibly being a high-level programming language such as PROLOG,

5 ontology — defines the vocabulary and meaning of the terms and concepts used in the content expression.

Level 1 provides structure for the dialogue between agents whereas levels 2—5 are contained in each FIPA ACL message. As with KQML, the FIPA ACL message comprises an s-expression. Figure 2 gives an example of a FIPA ACL message whereby the initiating agent `bt-agent` sends an `inform` message to `customer-agent` . The messaging information includes sending agent, receiving agent, the context for the message (`in-reply-to`) as well as the protocol (`fipa-contract-net`). The content of this message is a PROLOG statement using an ontology called `bt-auction` .

FIPA provides an optional content language, SL (semantic language), for expressing the content of ACL messages. FIPA SL is formally defined in the normative

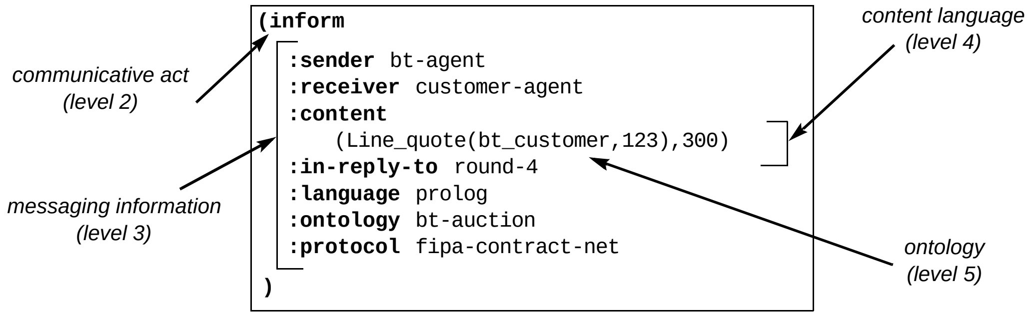

Fig 2 FIPA ACL message structure.

specification although its use is optional so that agent developers are not obliged to use SL for expressing their ACL content. In order to support uses of SL where the full expressiveness of SL is not required, subsets of SL (i.e. SL0, SL1, SL2) are defined. This allows for simpler agents and reduces the burden on agent developers who choose to use such content languages.

2.3 Communicative acts

FIPA defines a set of normative communicative acts. These represent a class of actions which are the building blocks of a dialogue between agents (e.g. `inform` and `request`). Each CA has a well-defined meaning associated with it that is independent of the overall content of the expression. FIPA defines both primitive and composite CAs. Primitive CAs represent the atomic actions and can be combined into more complex expressions, composite CAs. The standard set of CAs includes both primitive and composite ones (see Table 2), but allows the composition of new CAs providing the rules of composition expressed in the FIPA standard are followed. These rules ensure that the semantic integrity of the language is retained.

2.4 Agent interaction protocols

Agent interaction protocols define a sequence of messages which represent a complete conversation or dialogue between agents. This enables agents to anticipate each other's response according to a defined conversation pattern. A number of standard interaction protocols are specified in FIPA97. Figure 3 illustrates the FIPA-Request protocol which consists of an ordered exchange of communicative acts between two agents initiated by a `request` communicative act. The agent sending a `request` using this protocol can anticipate either a `not-understood`, `refuse` or `agree` response.

The FIPA specification does not restrict agent developers only to those protocols defined. Agent developers can adopt their own protocols or even allow a conversation structure to emerge from the interaction between agents. However, less complex agents might wish to restrict interactions to those protocols defined by FIPA. If an agent uses the standard FIPA protocol name, that agent has to abide by the protocol if it is to conform to FIPA97.

Table 2 Primitive and composite FIPA communicative acts.

Communicative act	Composite	Primitive
accept-proposal	✓	
agree	✓	
cancel	✓	
cfp*	✓	
confirm		✓
disconfirm		✓
failure	✓	
inform		✓
inform-if (macro act)	✓	
inform-ref (macro act)	✓	
not-understood	✓	
propose	✓	
query-if	✓	
query-ref	✓	
refuse	✓	
reject-proposal	✓	
request		✓
request-when	✓	
request-whenever	✓	
subscribe	✓	
* call for proposals		

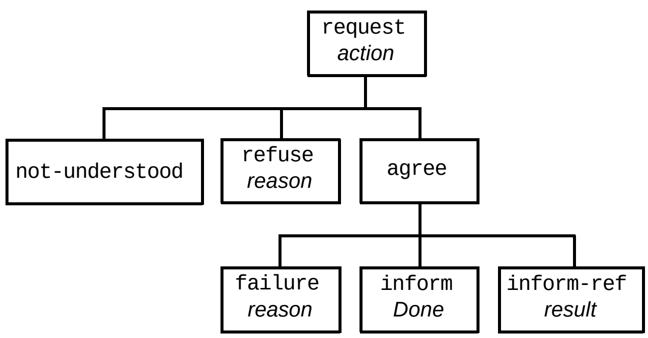

Fig 3 FIPA-request protocol.

2.5 Ontology services

Ontologies provide a means of representing and communicating knowledge about an application domain. This includes a vocabulary for expressing concepts in the application domain, integrity constraints and logical statements that describe the meaning of the terms. Interacting agents need to have a common ontology covering their domain of interaction if they are to understand each other. In order for FIPA to be an open standard the ontologies used by agents need to be explicitly stated and not remain implicit in the agent implementations. In both the agent management and agent-software integration domains, FIPA has defined related ontologies. An ontology reference service is currently being specified to support the storage and referencing of ontologies during agent interaction. Additionally, a top-level reference ontology based upon the ANSI top-level ontology is being considered. Version 1 of the FIPA ontology service is due in FIPA98.

3. Agent management

The main requirement for agent management was to specify an agent platform (AP) which had the following characteristics [11]:

- it should provide a messaging service that is reliable, orderly, and accurate,
- the agent should have a name enabling the message to be delivered to the correct destination,
- it should detect and report errors (e.g. failue to deliver a message) to the sending agent,
- it should minimise the reliance on a single interaction mechanism (i.e. HTTP, SMTP, GSM, etc),
- it should allow for facilities which support agent mobility and devices mobility,
- it should provide security services for messaging.

The FIPA agent management activity produced part 1 of the FIPA97 specification [10] which addresses all of the requirements except those for agent mobility and security. Specifications covering these requirements are currently being defined and will form parts 10 and 11 of FIPA98.

FIPA97 defines the infrastructure (the agent platform) (see Fig 4) for FIPA-compliant agents to enter an agent domain, advertise their services, locate other agents, and communicate with agents resident on other agent platforms.

The FIPA agent platform (AP) consists of the following mandatory components.

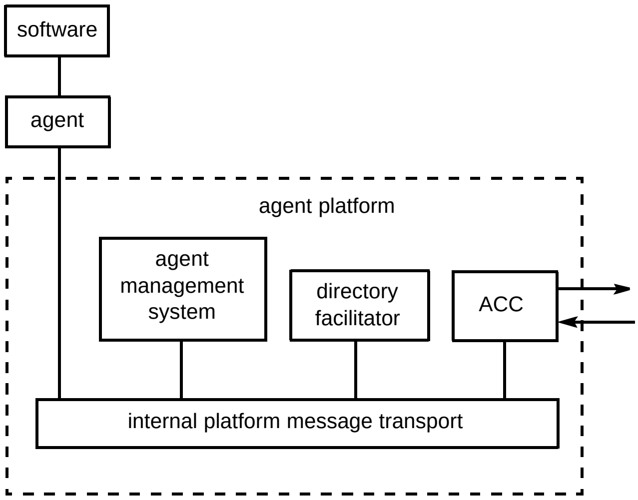

Fig 4 FIPA agent management reference model [9]

- Directory facilitator (DF)

  The DF is a special type of agent which provides ‘yellow pages’ services to other agents. Agents may register their services with the DF or query the DF for information on other agents. Membership of a particular DF defines a domain, which is an agent community that reflects the logical organisation of agents. These could be based upon organisational affiliation, market structure, or security domains. One AP can support multiple domains and likewise one domain can span multiple APs. DFs can register with each other forming a network of DFs allowing the escalation of queries between domains.

- Agent management system (AMS)

  The AMS provides an agent name service (ANS) whereby it maintains an index of all the agents which are currently registered with an AP. The index includes an agent’s unique name and their associated transport address for the AP. The agent’s name is composed of the agent’s home, agent platform address (i.e. the platform on which the agent was created) and a unique identifier within that platform [3]. The AMS is another type of agent which exerts supervisory control over access to and use of an AP. This includes creation of agents, deletion of agents, registration of agents on a the platform and overseeing the migration of agents to and from platforms.

- Agent communication channel (ACC):

  The ACC routes messages between APs. The ACC is the default communication method that connects all agents within an AP and between APs. In order to be FIPA-compliant, an AP must minimally support the Internet inter-orb protocol (IIOP). This is the default

[3] The agent’s unique name will remain with the agent throughout its life, irrespective of whether it migrates to other platforms or not.

inter-platform communication method which needs to be supported for interoperability between agent platforms. However, this does not preclude the use of other protocols in addition to the IIOP, nor does it require the use of IIOP within an agent platform.

The directory facilitator, agent management system, and agent communication channel are all types of agent and as such communicate via FIPA ACL. Agent management can be viewed as an application domain for intelligent agents and as such an agent management ontology has been defined to express the content of agent management communicative acts, e.g. an agent description.

Figure 5 provides an example of how a FIPA agent registers with a directory facilitator. The message structure is FIPA ACL. Registration uses the `request` communicative act and `fipa-request` protocol. The content expression is using SL0 and the FIPA agent-management ontology. Figure 5 also contains examples of the FIPA agent-naming convention.

The agent management specification provides facilities for mobile devices where there is intermittent connectivity to a network. These include the forwarding of messages to a proxy agent which can buffer messages when appropriate. The FIPA97 specification does not provide agent migration facilities (i.e. enable mobile agent code) which is currently being addressed in FIPA98 (see section 5). Similarly, FIPA97 has not addressed the important issue of agent security which again is being addressed during1998.

4. Agent software integration

The final normative part of FIPA97 [10] specifies a common way to encapsulate non-agent software and allow agents to access them as agents. A wrapper is specified which acts as an adaptor function between non-standard software and FIPA agent communication. A specialised agent called an agent resource broker (ARB) acts as specialised directory facility for non-agent software services that can be accessed via a FIPA wrapper. In Fig 6, wrapper agents enable access to non-agent software by mapping FIPA ACL into other languages.

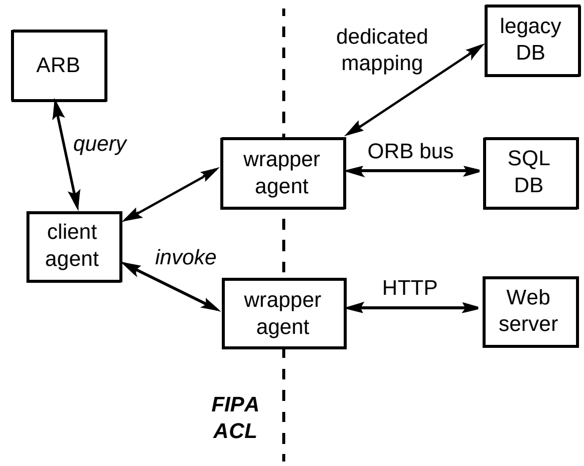

Fig 6 Agent — software integration.

5. FIPA and mobile agent standards

A related standardisation activity to FIPA is the Mobile Agent Systems Interoperability Facilities (MASIF)

```
(request
   :sender an-agent@iiop://fipa.org:50/acc
   :receiver a-df@iiop://fipa.org:50/acc
   :content
      (action a-df@iiop://fipa.org:50/acc
         (register
           (:df-description
           (:agent-name an-agent@iiop://fipa.org:50/acc)
           (:agent-service
              (:service-description
              (:service-type video-on-demand)
              (:service-ontology itut-vod)
              (:service-name vod-1)
              (:fixed-properties (genre sport))))
           (:interaction-protocols (fipa-request))
           (:ontology fipa-agent-management)
           (:address iiop://fipa.org/acc)
           (:ownership fipa.org)
           (:state active))))
   :language SL10
   :protocol fipa-request
   :ontology fipa-agent-management)
```

FIPA agent name

agent management content expression

ontology name

Fig 5 Agent management registration message.

specification for mobile agents. The MASIF specification which has recently been approved by the Object Management Group (OMG), defines the facilities necessary for software agents to migrate between agent platforms via CORBA IDL (Interface Definition Language) interfaces. MASIF uses the security, naming and life-cycle services offered by a CORBA platform.

FIPA has recognised the need to ensure that the FIPA and the MASIF specifications do not conflict. The main area where there is a need for concordance between FIPA and OMG MASIF is at the agent platform level. Presently, FIPA97 (Part 1) takes into account mobile devices (i.e. intermittent agent connectivity where the buffering of ACL messages is required by proxy agents in order for messages to be forwarded to agents which are temporarily unavailable). It does not explicitly provide support for the migration of agent code between agent platforms. However, intelligent agents might wish to migrate, and so FIPA is currently specifying facilities which will enable intelligent agents to migrate between agent platforms. MASIF provides one possible mechanism for the migration of agent code between platforms although it restricts developers to CORBA platforms.

FIPA is committed to retaining the agent developer's freedom to select a distributed computing platform of their choice and so does not wish to mandate the use of CORBA or any other specific platform. FIPA is specifying mobile agent facilities which support two types of mobility — the MASIF approach where migration facilities are embedded in the underlying platform (which could be CORBA based), and an alternative approach where the migrating agent can manage its own migration. This ensures that FIPA and OMG MASIF specifications [23] are concordant.

6. Open issues

6.1 Testing for compliance

FIPA, as with all standards, needs to be able to test for compliance and whether or not agent systems comply to the FIPA standard. However, problems exist in testing for compliance with such a semantically rich communication language as FIPA ACL. The FIPA ACL incorporates mentalistic notions of agent beliefs and desires which are difficult to validate based on external behaviour. Singh [24] outlines the problems associated with compliance testing of agent ACLs :

> 'To effectively test compliance, and especially to establish any violations, agents — human or artificial — must accumulate sub-traces of the executions (the 'smoking gun') that certify the violation.... The sub-traces must be derived from the agent's local view. Consequently, a research challenge is to design protocols for which the violation can be established with the relativised view of a single agent.'

The problems associated with compliance testing are being investigated by FIPA during 1998.

6.2 Validation of FIPA97

Apart from feedback from the production of the informative specifications, the FIPA97 specification is as yet largely untested. Organisations from both inside and outside FIPA are establishing trials to validate the specifications and provide feedback on their application. To facilitate this, FIPA issued informative specifications (parts 4—7) covering four application areas suited to agent technology. These serve both as examples of how to apply FIPA-compliant agent technology and as test cases for the normative part of the standards. They are intended to be the catalyst for field trials where the FIPA97 specifications can be validated and feedback provided for the refinement and improvement of FIPA97. Already, a number of validation initiatives have been established both as industrial collaborations and by individual companies, in the US, European Community and Asia-Pacific [4].

6.3 Implications on agent developers

FIPA has always set out to ensure that any intelligent agent standard does not impinge on the creativity of agent developers, and as such has only mandated the external behaviour of intelligent agents. However, the standardisation of agent interfaces at a knowledge level, as with FIPA ACL, has implications on the capabilities of FIPA-compliant agents. In particular, such agents must be able to communicate using FIPA ACL.

FIPA has been conscious of the need to specify the minimum necessary for interoperability between agents. A developers' guide is currently being produced to provide background material and an insight into the rationale behind many of the decisions made during the production of the FIPA specifications. This will also act as a guide to help developers use the FIPA97 specification.

7. Conclusions

FIPA97 provides a foundation for a standard to enable intelligent agent interoperation. It supports agent communication, agent management and agent/software integration. FIPA98 will refine the current specifications and extend them by adding facilities for agent mobility, security, human/agent interaction and ontology services.

[4] Under the European ACTS programme a number of collaborative projects have been established to investigate the use of agent technology, in particular in support of FIPA and MASIF standardisation efforts. Similarly, a Japanese industrial consortium is sharing experiences and collating feedback on FIPA specifications.

Agent standards are necessary if agent technology is to become commercially mature. FIPA has set itself a significant challenge in specifying a consortia standard where system interoperation is at such a knowledge-rich level. If FIPA achieves its goal and intelligent agent developers adopt the FIPA specifications, agent technology will represent a powerful software engineering paradigm for the next millennium.

Acknowledgements

The authors would like to acknowledge the contribution made by all those who have been actively involved in FIPA activities. They also express thanks to their colleagues in BT for comments on draft versions of this paper, in particular Paul Kearney. The views expressed in this paper are those of the authors and do not necessarily reflect the views of either BT or FIPA.

References

1 Busuoic M: 'Distributed cooperative agents for service management in communications networks', 11th IEE Teletraffics Symposium, Cambridge, England (1994).

2 Maes P: 'Agents that reduce work and information overload', Communications of the ACM, 37, No 7 (1994).

3 Davies N J, Weeks R and Revett M C: 'Information agents for the World Wide Web', in Nwana H and Azarmi N (Eds): 'Software agents and soft computing: towards enhancing machine intelligence,' Springer (1997).

4 Foner L and Crabtree B: 'Multi-agent matchmaking', in Nwana H and Azarmi A (Eds): 'Software agents and soft computing: towards enhancing machine intelligence', Springer (1997).

5 Fisk D: 'An application of social filtering to movie recommendation', in Nwana H and Azarmi N (Eds:) 'Software agents and soft computing: towards enhancing machine intelligence', Springer (1997).

6 O'Brien P D and Wiegand M E: 'Agents of change in business process management', in Nwana H and Azarmi N (Eds): 'Software agents and soft computing: towards enhancing machine intelligence', Springer (1997).

7 Ndumu D T, Collis J C and Nwana H S: 'Towards desktop personal travel agents', BT Technol J, 16, No 3, pp 69—78 (July 1998).

8 Nwana H S: 'Software agents: an overview', Knowledge Engineering Review, 11, No 3 (1996).

9 Nwana H and Wooldridge M: 'Software agent technologies', in Nwana H and Azarmi N (Eds): 'Software agents and soft computing: towards enhancing machine intelligence', Springer (1997).

10 FIPA97 Specification Part 1 Agent Management, Issue 1 (see http://drogo.cselt.it/fipa/) (1997).

11 FIPA97 Specification Part 2 Agent Communication, Issue 1 (see http://drogo.cselt.it/fipa/) (1997).

12 FIPA97 Specification Part 3 Agent Software Integration, Issue 1 (see http://drogo.cselt.it/fipa/) (1997).

13 Finin T et al: 'KQML as an agent communication language', in Bradshaw J (Ed): 'Software Agents', MIT Press (1997).

14 Neches R: 'Overview of the DARPA knowledge sharing effort', at http://www-ksl.stanford.edu/knowledge-sharing/papers/ksr-overview.html (1994).

15 Cohen P R and Levesque H J: 'Communicative actions for artificial agents', Proceedings of the First International Conference on Multi-Agent Systems, San Francisco, Cambridge, AAAI Press (1995).

16 Labrou Y and Finin T: 'A Semantic approach for KQML — a general purpose communication language for software agents', Proceedings of the 3rd International Conference on Information Knowledge Management (1994).

17 Labrou Y: 'Semantics for an agent communication language', PhD Thesis dissertation, University of Maryland, Baltimore (1996).

18 Sadek M D, Bretier P and Panaget F: 'Submission for standardisation of components of France Telcoms ARTIMIS technology', FIPA submission, Turin, Italy (1997).

19 Sadek M D: 'Logical task modelling for man-machine dialogue', Proceedings of AAAI'90, pp 970—975, Boston, MA (1990).

20 Sadek M D: 'Dialogue acts are rational plans', Proceedings of the ESCA/ETRW Workshop on the structure of multimodal dialogue, pp 1—29, Maratea, Italy (1991).

21 Sadek M D: 'A study in the logic of intention', Proceedings of the 3rd Conference on Principles of Knowledge Representation and Reasoning (KR'92), pp 462—473, Cambridge, MA (1992).

22 Sadek M D, Bretier P, Cadoret V, Cozannet A, Dupont P, Ferrieux A and Panaget F: 'A co-operative spoken dialogue system based on a rational agent model: a first implementation on the AGS application', Proceedings of the ESCA/ETR Workshop on Spoken Dialogue Systems: Theories and Applications, Vigso, Denmark (1995).

23 OMG TC Document: 'Mobile agent systems interoperability facilities specification', orbos/97-10-05 (1997).

24 Singh M: 'Principles of standardisation for agent communication', submission in response to FIPA 3rd Call for Proposals, Palo Alto (1998).

Paul O'Brien manages the Intelligent Business Systems Research group at BT Laboratories. He is researching the use of distributed intelligent agents for enterprise integration, process management and knowledge management. In the past, he has led numerous projects applying intelligent system technologies to service and network management.

He has a BA(Hons) in Philosophy and Computing from Hatfield Polytechnic, an MSc in Intelligent Knowledge Based Systems from Essex University, and an MSc(Econ) in Information Systems Management from the London School of Economics.

He currently chairs the Agent Management technical committee in FIPA (Foundation for Intelligent Physical Agents).

Richard Nicol is the manager of the 500-strong Applied Research and Technology Department at BT Laboratories.

He spent many years involved with video-coding research, technology, products and multimedia systems, work for which he was awarded the Martlesham Medal, the company's highest accolade for technical achievement.

For 3 years he ran BT's research activities with the Universities, and has advised numerous Governmental and Academic organisations on telecommunications research strategy.

He is a Chartered Engineer, a Fellow of the Institution of Electrical Engineers, a founding Director of FIPA and a visiting Professor at the University of Essex.